Phys 6268 Midterm Review

Subject Matter:

Strogatz chapter	Lecture	lec: #	Problem/Topic
—— 1D flows ——			
1. Overview	Introduction	1	Geometric thinking, equilibria
2. Flows on the line	Flows in 1D	2	Fixed points Stability linear stability analysis Existence/Uniqueness Impossibility of oscillations
	Numerical Soln. of ODEs	3	Potentials numerical methods
3. Bifurcations	Bifurcations in 1D	4	Defining bifurcations Saddle-node bifurcation transcritical bifurcation Laser threshold problem
	" (2)	5	Pitchfork bifurcation Overdamped bead on rotating loop Imperfect bifurcations catastrophe theory
4. Flows on the circle	Flows on a Circle	6	Uniform oscillator non uniform oscillator Overdamped pendulum synchronization Superconducting Josephson junction
—— 2-D flows ——			
5. Linear Systems	2-D systems	7	generalizing linear systems Classification of linear systems
6. Phase Plane	Phase Plane Analysis	8	Phase portraits Existence, uniqueness, topological consequences Fixed points & Linearization Nullclines
	Conservative Systems	9	Conservative systems Reversible systems Pendulum dynamics

CMH

Nonlinear Dynamics Midterm Review

Strogatz notes [pg. no.]

CH2: Flows on the Line

[8] • In general, autonomous (linear) ODE systems can be written in the form

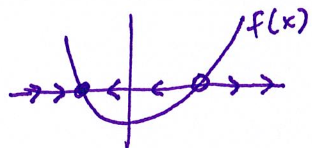
$$\dot{x}_1 = f_1(x_1, \dots, x_n) \quad \vdots \quad \dot{x}_n = f_n(x_1, \dots, x_n) \quad \mapsto \quad \dot{\vec{x}} = \vec{f}_n(\vec{x}) \quad \text{or} \quad \dot{\vec{x}} = \underline{\underline{A}} \cdot \vec{x} \text{ (for linear ones)}$$

– in the non-autonomous case, we introduce $x_{n+1} = t$, such that

$$\dot{x}_1 = f_1(x_1, \dots, x_n) \quad \vdots \quad \dot{x}_n = f_n(x_1, \dots, x_n) \quad \dot{x}_{n+1} = 1$$

since $\dot{x}_{n+1} = 1$, $x_{n+1} = t$, and we circumvent non autonomy to write the system in the same form

[18] • For the dynamical system $\dot{x} = f(x)$, the phase portrait of the system is



- intersections of $f(x^*) = 0$ occur at fixed points x^*
- for $\dot{x} = f(x) + g(x)$, analyze $f(x)$, $-g(x)$ on the same axes and look for intersections ($\dot{x} = 0 \Leftrightarrow f(x) = -g(x)$)

• Linear Stability Analysis: linearization about x^* in 1D systems

[24] • let x^* be a fixed point, $\eta(t) = x(t) - x^*$ is a small perturbation away

$$\Rightarrow \dot{\eta} = \frac{d}{dt}(x - x^*) = \dot{x} - 0 = \dot{x} \Rightarrow \dot{\eta} = f(x^* + \eta) \approx f(x^*) + \eta f'(x^*)$$

then neglect $O(\eta^2)$... $\Rightarrow \dot{\eta} \approx \eta f'(x^*)$

• if $f'(x^*) \neq 0$, η grows exponentially if $f'(x^*) > 0$, and decays when $f'(x^*) < 0$

• hence, characteristic timescale of this behavior is just $\tau = \frac{1}{|f'(x^*)|}$

↳ characteristic timescale of this behavior is just

• if $f'(x^*) = 0$, then $O(\eta^2)$ analysis is required! (graph the curve!)

• Existence & Uniqueness

[27] • consider the system $\{\dot{x} = f(x),\ x(t=0) = x_0\}$

• if $f(x)$ and $f'(x)$ are continuous in an open interval R, and $x_0 \in R$, then the IVP has a solution $x(t)$ on some time interval $[-\tau, \tau]$

• This solution $x(t)$ is unique. However, not guaranteed valid for all time!

• Impossibility of Oscillations in 1D systems

[29] • in 1D systems, approach to equilibrium is always monotonic

• topological consequence → if line became periodic domain, like a circle, then periodic solutions are permitted

• mechanical analog: overdamped systems

• Potentials of a Vector Field

• we define a potential $V(x)$ for the system: $f(x) = -\frac{dV}{dx}$

• the potential $V(x)$ must decrease along trajectories $x(t)$.

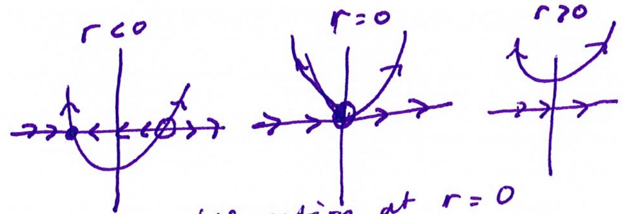
• $\frac{dV}{dx} = 0$ @ x^*, since there is no motion at a fixed point

• choice of integration constant for convenience (only dV or ΔV matters)

CH3: Bifurcations

[45] • Bifurcations are qualitative changes in dynamics due to a system/parameter change

• Saddle-node Bifurcation

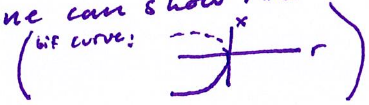


[46]
- prototypical ex: $\dot{x} = r + x^2$
- this admits the phase portraits:
- also: fold, turning point, blue sky

bifurcation at $r=0$

- in 1-D, if $\dot{x} = g(x,r) + h(x)$ then the bifurcation occurs where $g(x,r) = h(x)$ and $g'(x,r) = h'(x)$ (curves tangentially intersect)
- by Taylor expanding, (about $(r-r_c)$, $(x-x^*)$), one can show the normal form: $\dot{x} = a(r - r_c) + b(x-x^*)^2$

(bif curve:



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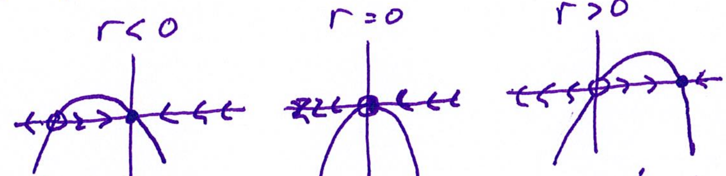
• Transcritical Bifurcation

[51]
- normal form: $\dot{x} = rx - x^2$
- fixed point always exists, but encounters an exchange of stability



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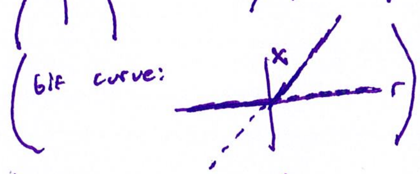

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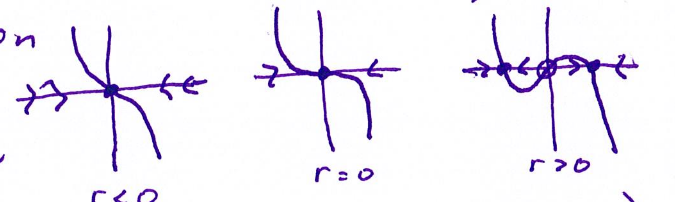
• Pitchfork Bifurcation

[56]
- Supercritical Pitchfork Bifurcation
  - normal form: $\dot{x} = rx - x^3$
  - @ $r=0$, still fixed point $x^* = 0$, but linearization vanishes: much slower decay than exponential
    ↳ Critical slowing down
  - Fixed points in triplet (see normal form)



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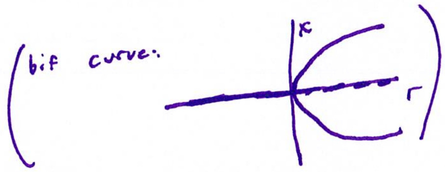

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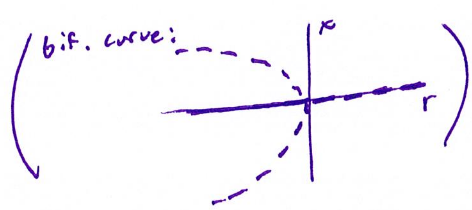
- Subcritical Pitchfork Bifurcation
  - normal form: $\dot{x} = rx + x^3$
  - pitchfork is inverted, only unstable $x^* \neq 0$
  - origin still always a fixed point

(bif. curve:



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• Imperfections & Catastrophes

[70]
- Consider $\dot{x} = h + rx - x^3$ → h is the imperfection parameter, breaking symmetry
- Can consider parameter space vs. x to visualize the "folding" or catastrophe in parameter space

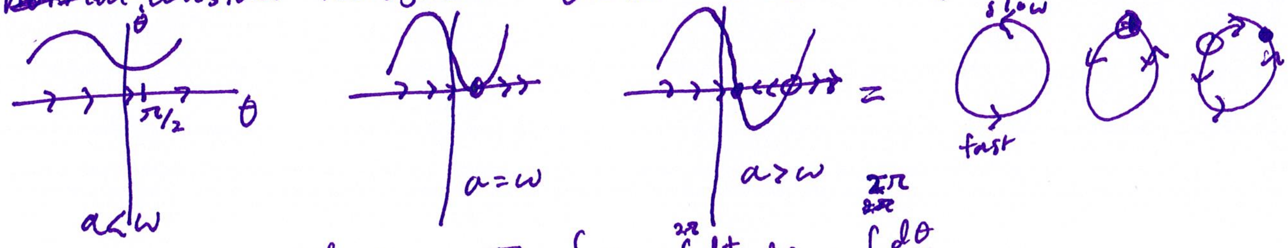
CH4: Flows on a Circle

• Uniform Oscillator

[97]
- phase θ changes uniformly: $\dot{\theta} = \omega$ ($\Rightarrow \theta(t) = \omega t + \theta_0$)
- period is given by $T = \frac{2\pi}{\omega}$
- For 2 uniform oscillators $\dot{\theta}_1 = \omega_1$, $\dot{\theta}_2 = \omega_2$, beats occur at the time interval: $T_{beat} = \left(\frac{1}{T_1} - \frac{1}{T_2}\right)^{-1}$

• Non-Uniform Oscillator

[98] • ~~Consider~~ Consider the system $\dot{\theta} = \omega - a\sin\theta$ (ω: mean, a: amplitude)



• Oscillation period is $T = \int dt = \int_0^{2\pi} \frac{dt}{d\theta}\, d\theta = \int_0^{2\pi} \frac{d\theta}{\dot{\theta}}$

• overdamped system pendulum ↔ Josephson junctions

2 Dimensional Flows

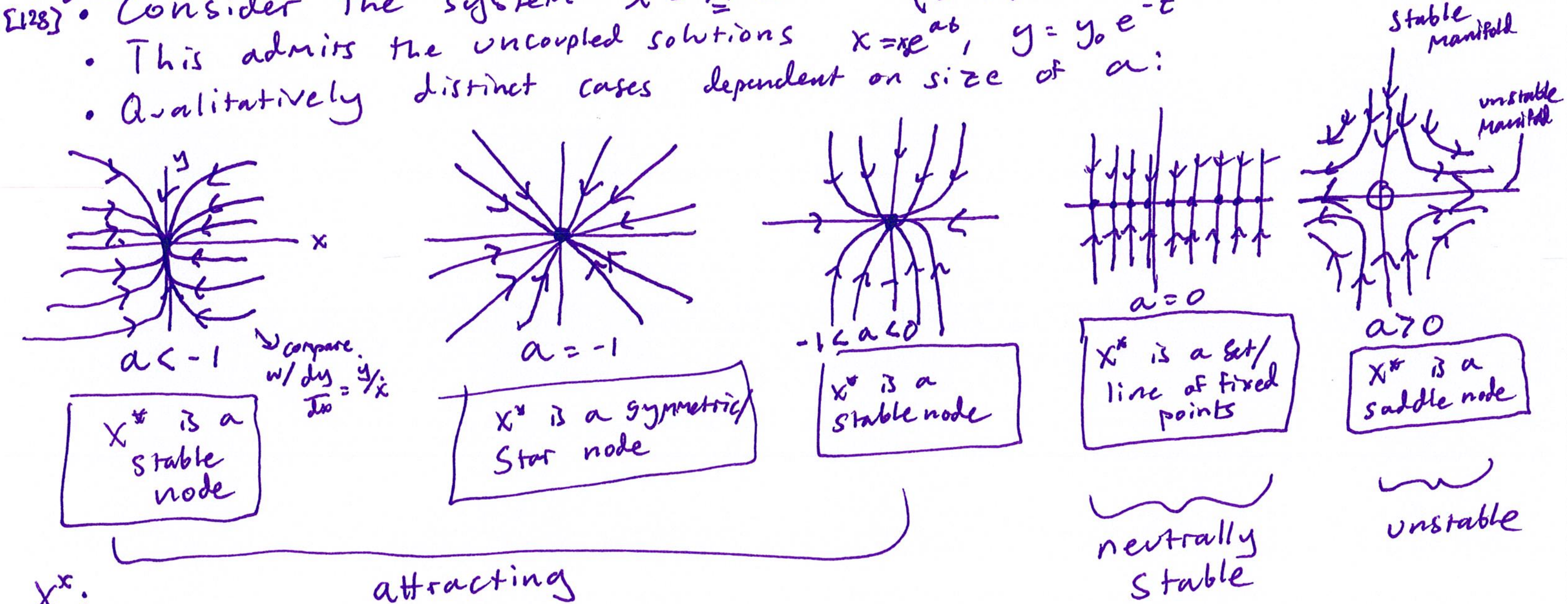
CH5: Linear Systems

• Example

[128] • Consider the system $\dot{\vec{x}} = \underline{\underline{A}} \cdot \vec{x} = \begin{pmatrix} a & 0 \\ 0 & -1 \end{pmatrix} \cdot \vec{x} \longrightarrow \dot{x} = ax;\ \dot{y} = -y$

• This admits the uncoupled solutions $x = x_0 e^{at}$, $y = y_0 e^{-t}$

• Qualitatively distinct cases dependent on size of a:



• Classification of Linear Systems

[131] • In general, $\dot{\vec{x}} = \underline{\underline{A}} \cdot \vec{x}$; $\underline{\underline{A}}^{(2\times 2)}$ admits characteristic eqn. for eigenvalues:

$$\lambda^2 - (\mathrm{Tr}\,\underline{\underline{A}})\lambda + (\det \underline{\underline{A}}) = 0 \Rightarrow \left[\lambda_\pm = \frac{\mathrm{Tr}\,\underline{\underline{A}}}{2} \pm \frac{1}{2}\sqrt{(\mathrm{Tr}\,\underline{\underline{A}})^2 - 4\det \underline{\underline{A}}} \right]$$

• Then, solutions space is spanned by the eigensolutions:

$$\vec{x}_\pm(t) = e^{\lambda_\pm t} \cdot \vec{V}_\pm \quad \text{where } \vec{V}_\pm \text{ are eigenvectors}$$

$$\Rightarrow \vec{X}(t) = c_+ e^{\lambda_+ t}\, \vec{V}_+ + c_- e^{\lambda_- t}\, \vec{V}_-$$

• The classification can be summarized

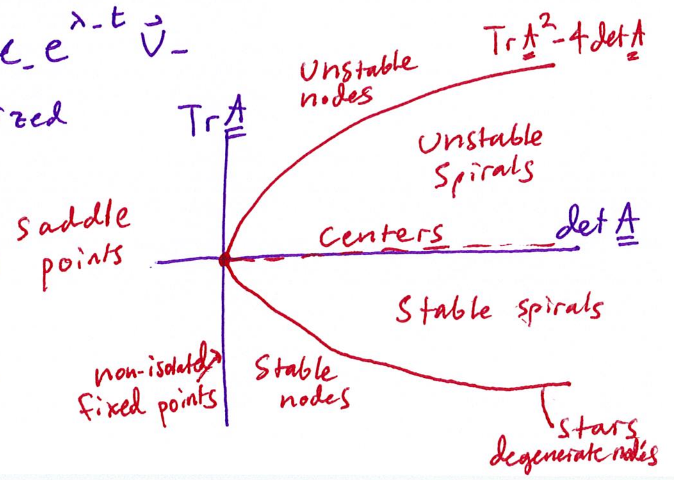
→ $\det \underline{\underline{A}} < 0$: e-vals real w/ opposite signs → saddle

→ $\det \underline{\underline{A}} > 0$: either real w/ same sign (node) or complex conjugate (spirals & centers) for ($\mathrm{Tr}\,\underline{\underline{A}}^2 > 4\det\underline{\underline{A}}$, $\mathrm{Tr}\,\underline{\underline{A}}^2 < 4\det\underline{\underline{A}}$)

→ $\mathrm{Tr}\,\underline{\underline{A}}^2 - 4\det\underline{\underline{A}} = 0$: Star nodes / degenerate nodes

→ stability of nodes/spirals: $\begin{cases} \mathrm{Tr}\,\underline{\underline{A}} < 0 \Rightarrow \text{stable} \\ \mathrm{Tr}\,\underline{\underline{A}} = 0 \Rightarrow \text{neutral center} \\ \mathrm{Tr}\,\underline{\underline{A}} > 0 \Rightarrow \text{unstable} \end{cases}$

→ $\det\underline{\underline{A}} = 0$: non-unique FP!



CH6: Phase Plane Analysis

- Phase Portraits

[146]
  - general form of our systems: $\dot{\vec{x}} = \vec{f}(\vec{x})$
  - Primary features of phase portraits
    - fixed points
    - closed orbits
    - Stability of points/orbits

- Existence & Uniqueness

[150]
  - existence/uniqueness guaranteed if $\vec{f}$ is continuously differentiable
  - corollary: distinct trajectories never intersect
  - implies that trajectories inside of closed orbits either go to orbit or fixed point ($\approx$ Poincaré-Bendixson thm)

- Linearization

[151]
  - consider the system $\begin{cases} \dot{x} = f(x,y) \\ \dot{y} = g(x,y) \end{cases}$ w/ x^*, y^* a fixed point. Then,

    let $u = x - x^*$, $v = y - y^*$, $\dot{u} = \dot{x}$, $\dot{y} = \dot{v}$, so we have

    $\dot{u} \approx u \frac{\partial f}{\partial x}(x^*, y^*) + v \frac{\partial f}{\partial y}(x^*, y^*)$; $\dot{v} = u \frac{\partial g}{\partial x}(x^*, y^*) + v \frac{\partial g}{\partial y}(x^*, y^*)$

  - or compactly, for (x,y) near (x^*, y^*), the system is roughly

$$\dot{\vec{x}} = \underbrace{\begin{pmatrix} \frac{\partial f}{\partial x}(x^*, y^*) & \frac{\partial f}{\partial y}(x^*, y^*) \\ \frac{\partial g}{\partial x}(x^*, y^*) & \frac{\partial g}{\partial y}(x^*, y^*) \end{pmatrix}}_{\text{Matrix of Stability}} \cdot (\vec{x} - \vec{x}^*)$$

(≠ Jacobian)

  - Linearization breaks down if the fixed point is a borderline case: (centers, degenerate nodes, stars, non-isolated FP's)

  - Stability of Fixed Points:

Robust Cases

- repellers (sources): $\lambda_\pm > 0$
- attractors (sinks): $\lambda_\pm < 0$
- saddle : $\lambda_- < 0 < \lambda_+$

(for real part of λ, Reλ)

Marginal Cases

- Centers: $\lambda_\pm$ are pure imaginary
- high-order/non-isolated: one λ is 0, λ_+ or $\lambda_- = 0$

  - if Re$\lambda \neq 0$ for $\lambda_\pm$, then the fixed point is hyperbolic

- Conservative systems

[160]
  - Systems for which a potential exists are conservative, and thus admit a conserved quantity
  - for $\dot{\vec{x}} = \vec{f}(\vec{x})$, a conserved quantity $E(\vec{x})$ is real-valued, continuous, and constant along trajectories
  - homoclinic orbit: trajectories that start/end @ the same fixed point $\vec{x}^*$
  - heteroclinic orbit: orbits joining one or more saddle points
  - Reversible systems are invariant under $(x, y, t) \longmapsto (+x, -y, -t)$ or $\vec{x} \to \vec{R}(\vec{x})$ s.t. $\vec{R}^2(\vec{x}) = \vec{x}$